Volume of prisms

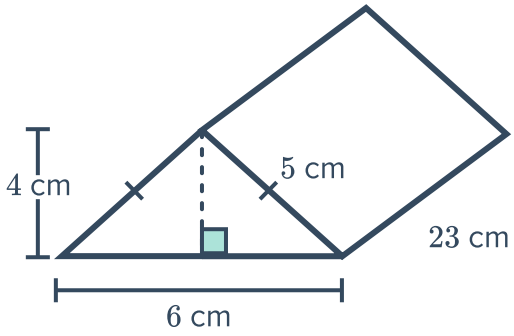
1 For each of the following prisms:

i State the shape of the prism's base.

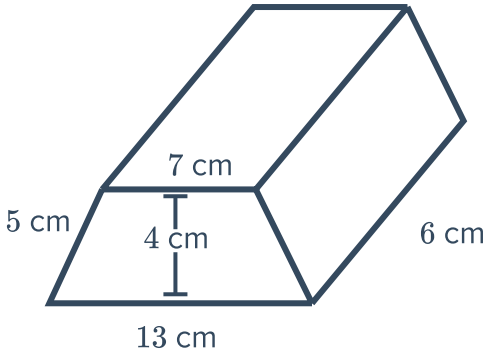
ii Find the area of the prism's base.

iii Find the volume of the prism.

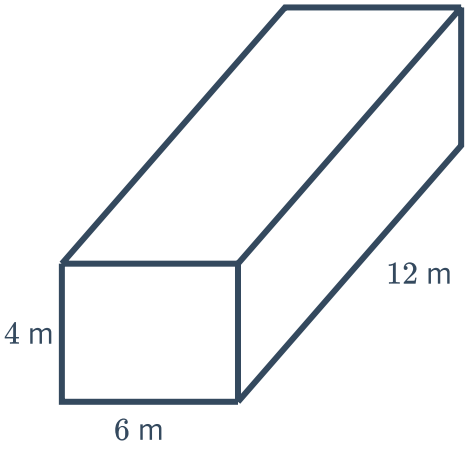
a



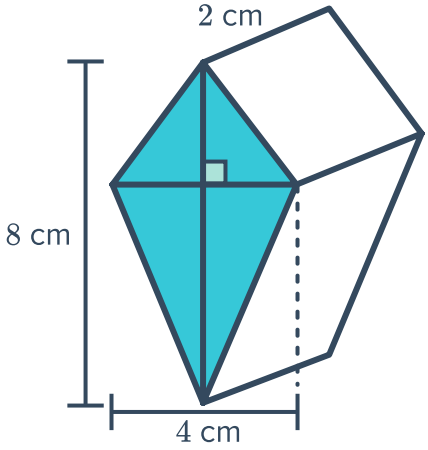
b



c



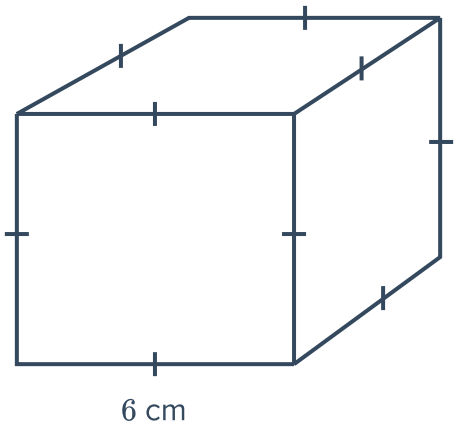
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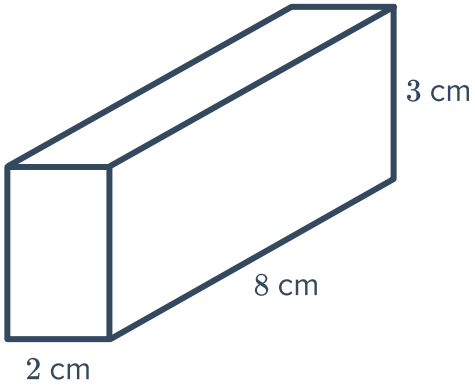
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2 Find the volume of the following prisms:

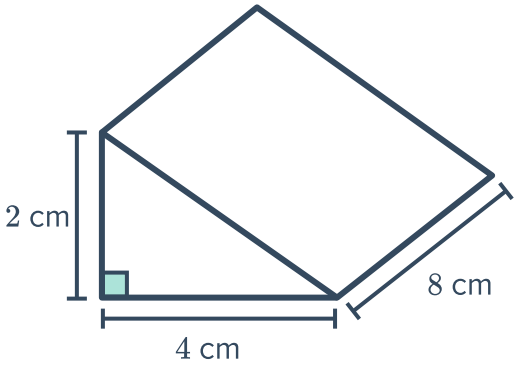
a



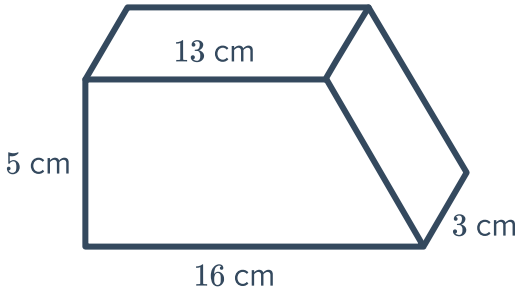
b



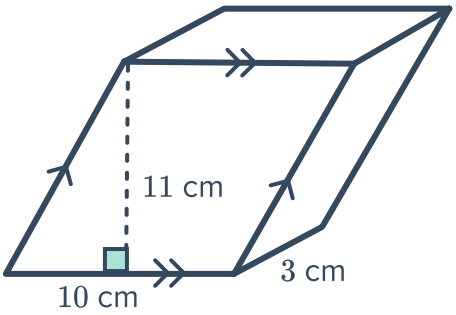
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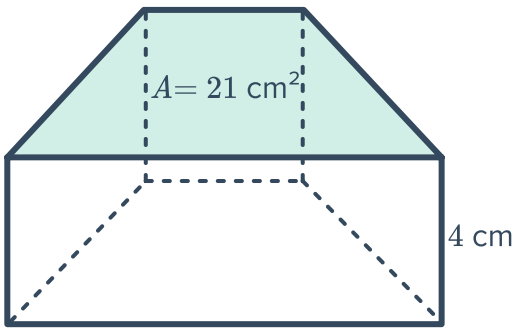
d



e

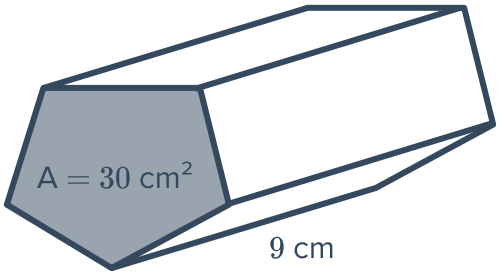


f



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g



3 A triangular prism with a base area of $20\,\text{cm}^2$ and a height of $10\,\text{cm}$. Find its volume.

4 An octagonal prism with a base area of $80\,\text{mm}^2$ and a height of $120\,\text{mm}$. Find its volume.

Volume of cylinders

5 Rochelle notices that the base of a cylinder is always a circle. To save working out time, Rochelle decides to combine the area formula $A = \pi r^2$ with the volume formula $V = Ah$.

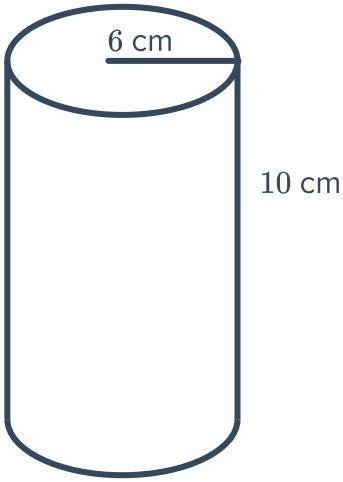
By substituting the area formula into the volume formula, state the formula Rochelle gets for the volume of a cylinder.

6 Consider the solid shown in the diagram:

a State the shape of the base of this solid.

b Find the exact area of the solid's base.

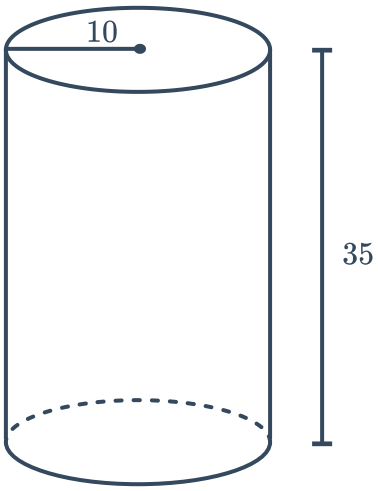
c Find the exact volume of the solid.



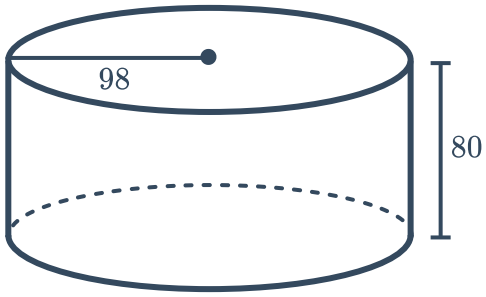
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7 Find the volume of the following solids, rounding your answers to one decimal place:

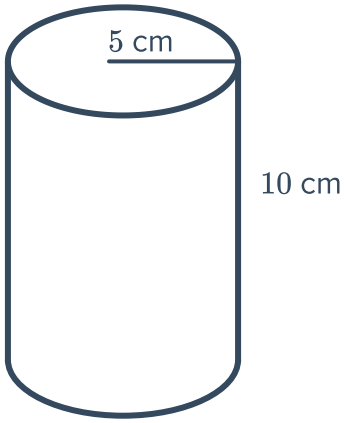
a



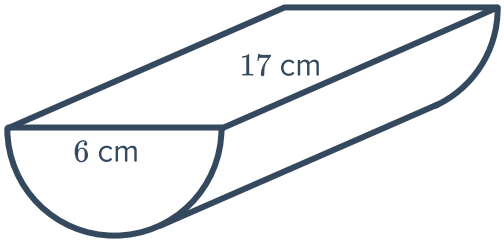
b



c



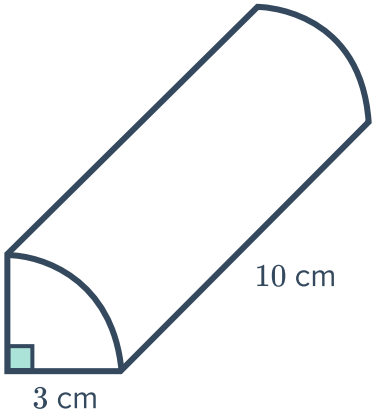
d



e

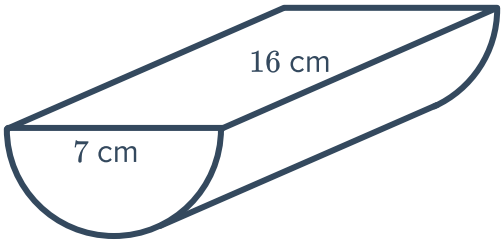


f

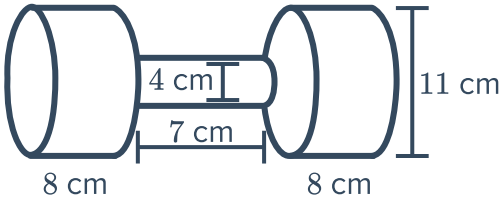


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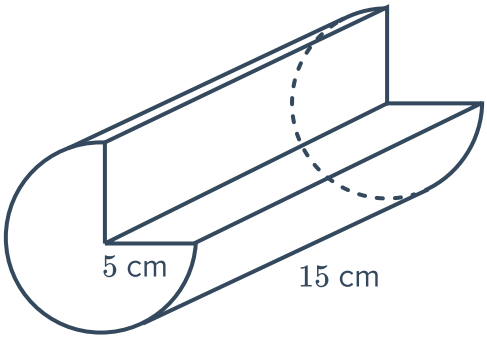
g



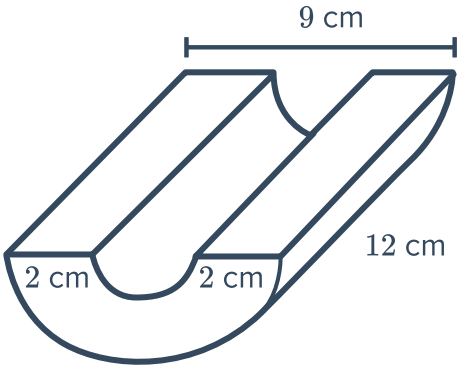
h



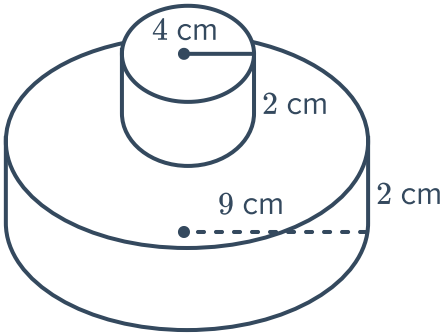
i



j



k



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Volume of compound solids

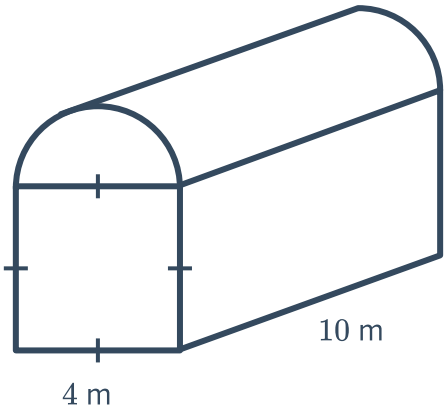
8 For each of the following compound solids:

i State the basic shapes used to make this compound shape.

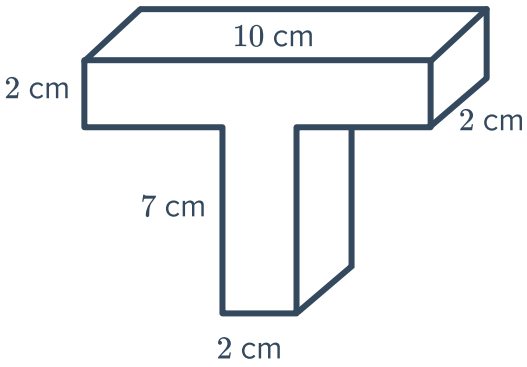
ii Find the area of the solid's base, rounded to three decimal places.

iii Find the volume of the solid, rounded to two decimal places.

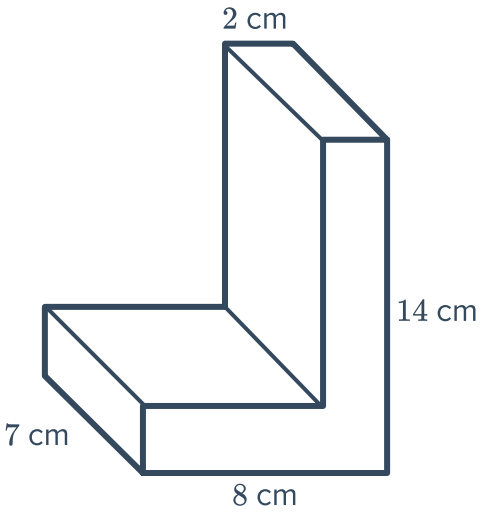
a



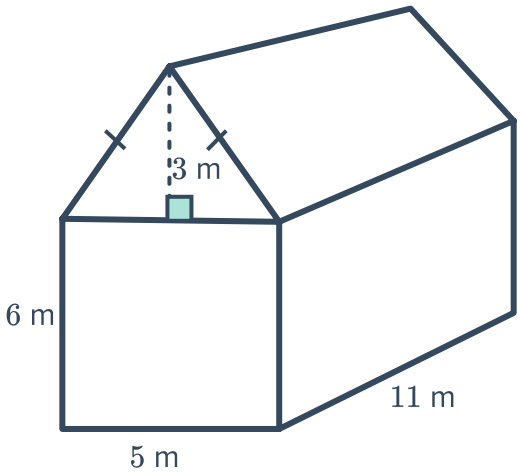
b



c



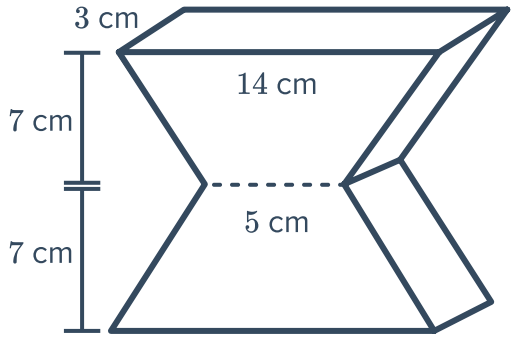
d



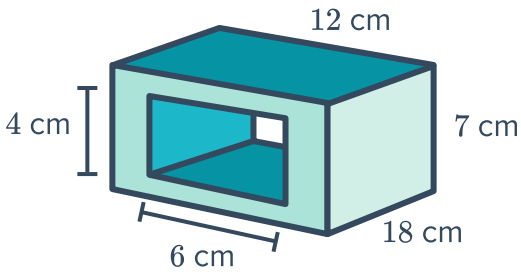
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9 Find the volume of the following compound solids. Round your answers to two decimal places where necessary.

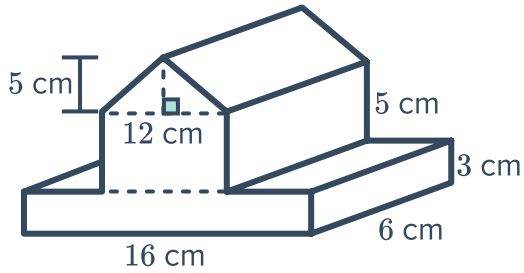
a



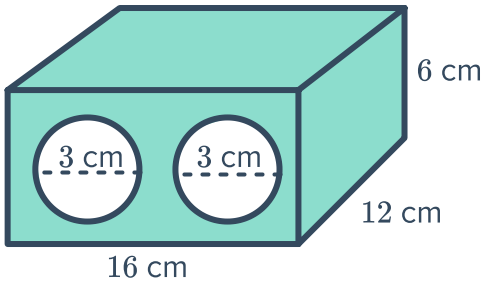
b



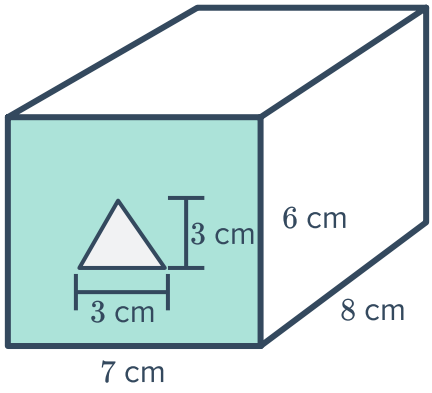
c



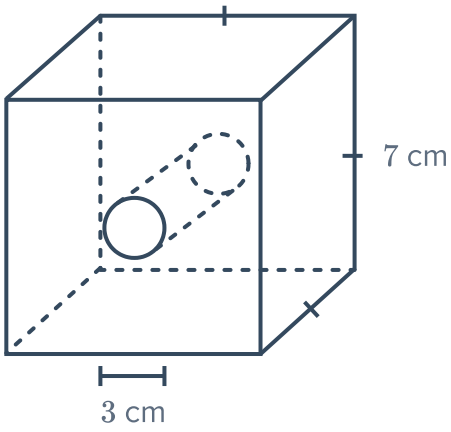
d



e



f

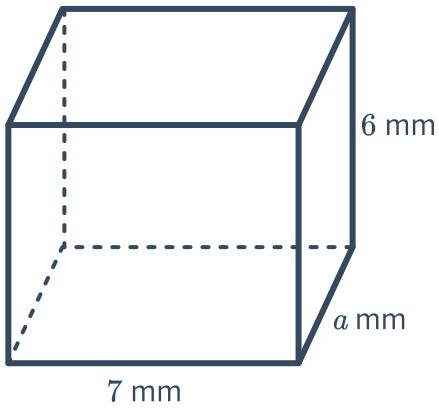


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Unknown side given the volume

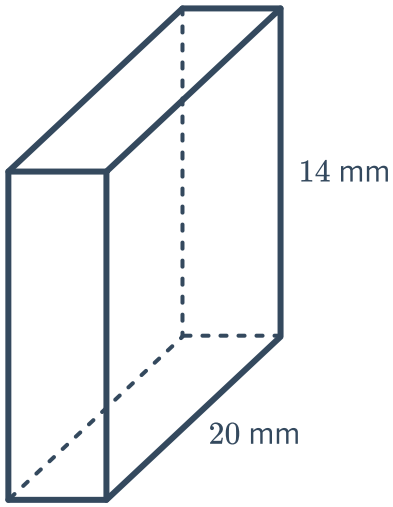
10 The following rectangular prism has a volume of $168\ \text{mm}^3$:

Find the length, a, of the rectangular prism.



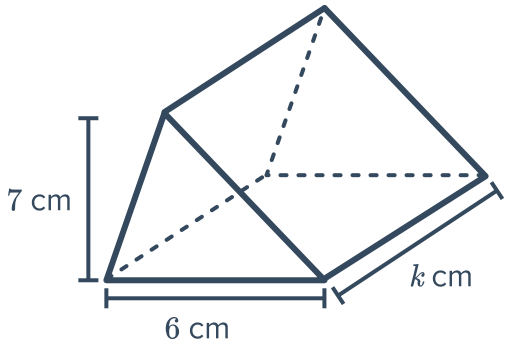
11 The following rectangular prism has a volume of $1680\ \text{mm}^3$:

Find the width of the rectangular prism in millimetres.



12 The volume of the triangular prism shown is $231\ \text{cm}^3$.

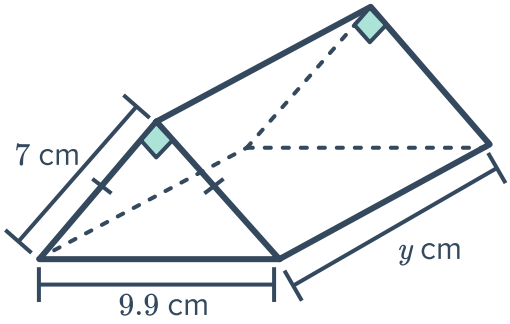
Find the value of k.



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13 The volume of the triangular prism shown is $247.45\ \text{cm}^3$.

Find the value of y to one decimal place.



14 A prism has a volume of $770\ \text{cm}^3$. Given it has a height of $11\ \text{cm}$, find the base area of the prism.

15 A prism has a volume of $990\ \text{cm}^3$. Given it has a base area of $110\ \text{cm}^2$, find the height of the prism.

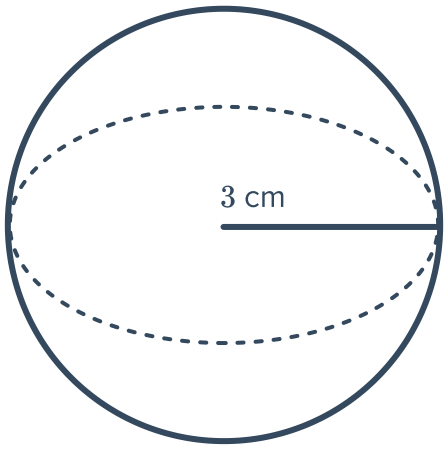
16 Find the side length of a cube with volume $27\ \text{cm}^3$.

17 Find the length of the rectangular prism with volume $162\ \text{cm}^3$, width $6\ \text{cm}$ and height $3\ \text{cm}$.

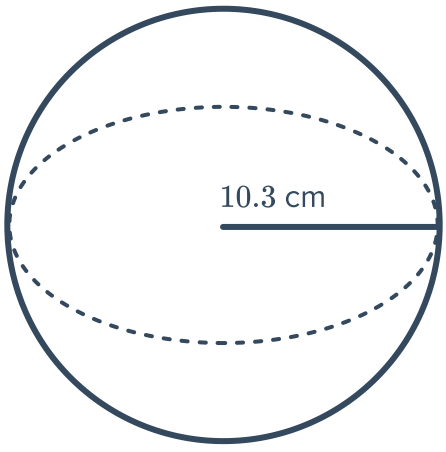
Volume of spheres

18 Find the volume of the following spheres. Round your answers to two decimal places.

a

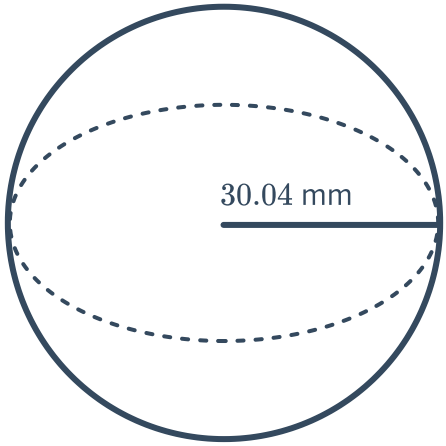


b

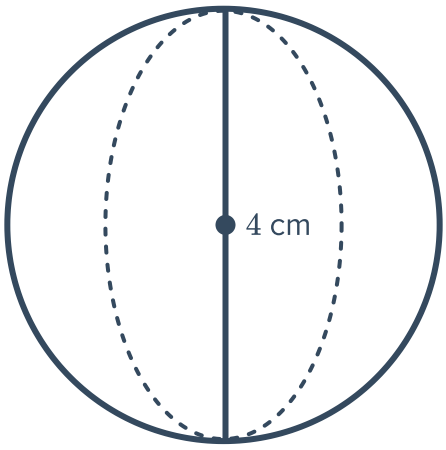


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c



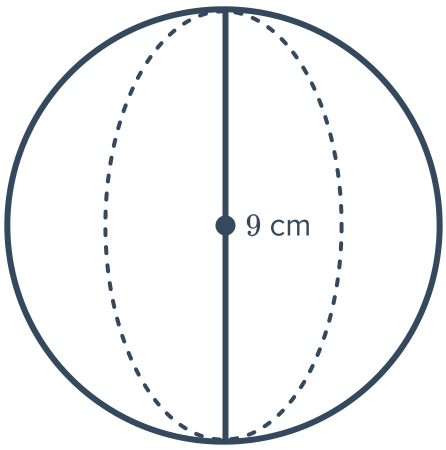
d



19 A student was calculating the volume of the sphere shown. The working is given below:

$$\begin{aligned} \text{Step 1} \quad V &= \frac{4}{3}\pi \times 9^3 \\ \text{Step 2} \quad &= \frac{2916}{3}\pi \\ \text{Step 3} \quad &\approx 3053.63 \text{ cm}^3 \end{aligned}$$

Explain what the student's mistake was, and find the actual volume of the sphere.



20 Find the volume of the following spheres. Round your answer to three decimal places.

a A sphere with a radius of 5.8 mm.

b A sphere with a diameter of 17 cm.

c A sphere with a radius of 0.9 m.

d A sphere with a diameter of $\frac{7}{8}$ cm.

Applications

21 A garden bed is 5 m in length, 2 m in width and 20 cm in height.

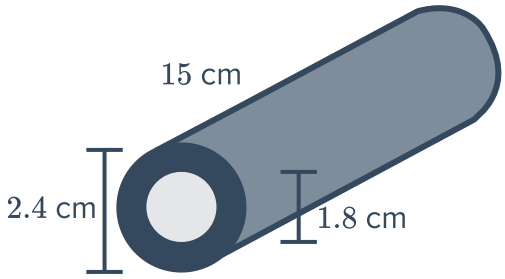
Find the volume of soil in cubic metres that will be needed to fill up the garden bed.

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22 A hollow cylindrical pipe has the dimensions shown below:

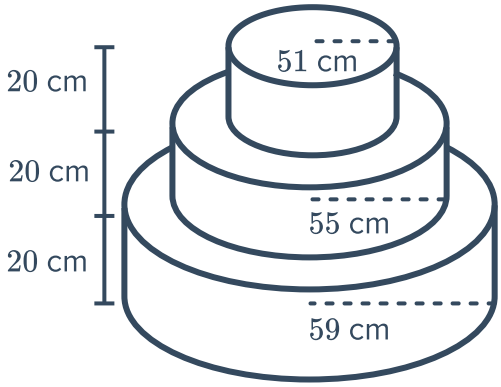
a Calculate the volume of the pipe shown, correct to two decimal places.

b Calculate the weight of the pipe if $1\ \text{cm}^3$ of metal weighs $5.7\ \text{g}$, rounding your answer to one decimal place.



23 A wedding cake with three tiers is shown. The layers have radii of $51\ \text{cm}$, $55\ \text{cm}$ and $59\ \text{cm}$. If each layer is $20\ \text{cm}$ high, calculate the total volume of the cake in cubic metres.

Round your answer to two decimal places.

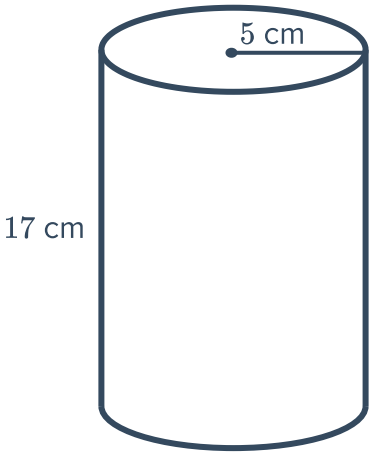


24 Jack's mother told him to drink 3 large bottles of water each day. She gave him a cylindrical bottle with height $17\ \text{cm}$ and radius $5\ \text{cm}$.

a Find the volume of the bottle. Round your answer to two decimal places.

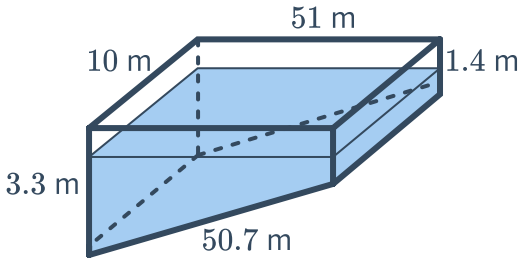
b Assuming that he drinks 3 full bottles as his mother suggested, calculate the volume of water Jack drinks each day. Round your answer to two decimal places.

c If Jack follows this drinking routine for a week, how many litres of water would he drink altogether? Round your answer to the nearest litre.



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25 A swimming pool has the shape of a trapezoidal prism as shown in the diagram:



a Find the volume of the pool in cubic metres.

b If the pool is three-quarters full what is the volume of the non filled space of the pool.

c If the distance of the water level to the top of the pool is h m when it is is three-quarters full, then find h.

26 A box of tissues is in the shape of a rectangular prism. It has a length of 39 cm, a width of 19 cm and a height of 11 cm.

a Find the volume of the box.

b If the shelf at the supermarket is 95 cm long and has a depth of 40 cm, find the maximum number of tissue boxes that can fit on the shelf.

27 A special refrigerator is used to store medical samples and has dimensions 18 cm by 70 cm by 12 cm. The samples are stored in small containers that have dimensions 20 mm $\times$ 50 mm $\times$ 20 mm.

Assuming both the refrigerator and the sample containers are rectangular prisms,

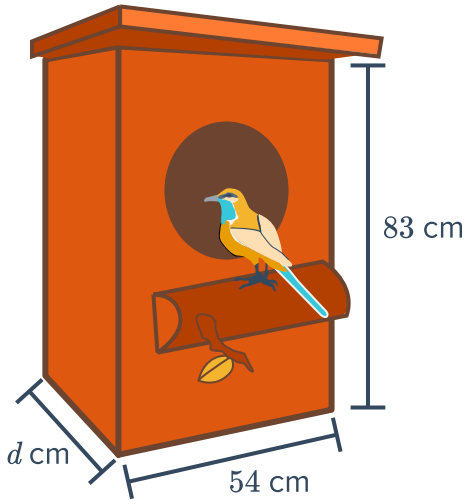
a Find the dimensions of the sample containers in centimetres.

b Find the volume of a sample container.

c Find the volume of the fridge.

d How many containers can be stored in the fridge?

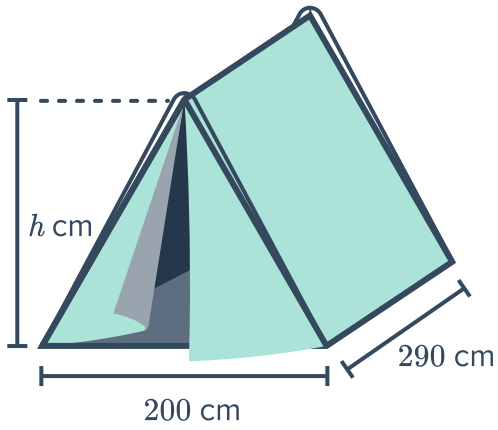
28 This nesting box needs to have a volume of $129\,978$ cm^3, a height of 83 cm and a width of 54 cm. Find the depth, d, of the box.



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29 The volume of the following tent is $4.64\ \text{m}^3$.

Find the height, h, of the tent.



30 Find the volume of a bowling ball with a radius of $10.9\ \text{cm}$. Round your answer to three decimal places.

31 How many whole lead balls with a diameter of $0.5\ \text{cm}$ can be made from the amount of lead in a ball with a diameter of $10\ \text{cm}$?

32 How many cubic centimetres of gas are necessary to inflate a spherical balloon to a diameter of $60\ \text{cm}$? Round your answer to the nearest cubic centimetre.

33 The planet Jupiter has a radius of $69\,911\ \text{km}$, and planet Mercury has a radius of $2439.7\ \text{km}$. How many times bigger is the volume of Jupiter than Mercury? Assume that both planets are spheres. Round your answer to one decimal place.

34 The planet Mars has a radius of $3400\ \text{km}$. What is the volume of Mars? Write your answer in scientific notation to three decimal places.

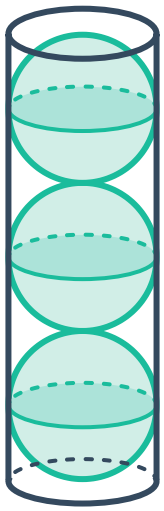
35 A cubic box has a volume of $34\,500\ \text{cm}^3$.

a Find the side length of the box, correct to 4 decimal places.

b Using the answer from part (a), find the volume of the largest ball that can fit inside the box. Write your answer correct to the nearest cm^3.

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36 Three spheres of radius $4\ \text{cm}$ fit perfectly inside a cylindrical tube so that the height of the three spheres is equal to the height of the tube, and the width of each sphere equals the width of the tube:



a Find the total volume of the three spheres. Round your answer to one decimal place.

b Find the volume of the tube. Round your answer to one decimal place.

c Calculate the percentage of the space inside the tube that is not taken up by the spheres. Round your answer to the nearest whole percent.

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